

Garrett Brownlee
Dr. Kieth Sanford
PSY 4400 01
11/15/2025

Project #4: Regression

Step #5 (w/ regression equation #1):

```
Console Terminal Background Jobs
R 4.5.1 · ~/Brownlee/
> listWPCC <- c('pbeh_72.w', 'pbeh_74.w', 'pbeh_76.w', 'pbeh_81.w', 'pbeh_83.w', 'pbeh_85.w', 'pbeh_87.w')
> proj4$w_prtnrcollabcomm_GB <- rowSums(proj4[, listWPCC])
> listWPE <- c('exp_152.w', 'exp_153.w', 'exp_154.w', 'exp_155.w')
> proj4$w_posexpectns_GB <- rowSums(proj4[, listWPE])
> listWOSE <- c('emo_52.w', 'emo_55.w', 'emo_62.w', 'emo_65.w')
> proj4$w_ownsoftemo_GB <- rowSums(proj4[, listWOSE])
> library(psych)
> listHres <- c('db_Resolve.H', 'w_prtnrcollabcomm_GB', 'w_posexpectns_GB', 'w_ownsoftemo_GB')
> corr.test(proj4[, listHres])
Call:corr.test(x = proj4[, listHres])
Correlation matrix
      db_Resolve.H w_prtnrcollabcomm_GB
db_Resolve.H      1.00                0.24
w_prtnrcollabcomm_GB 0.24                1.00
w_posexpectns_GB    0.41                0.46
w_ownsoftemo_GB     -0.08               -0.19
      w_posexpectns_GB w_ownsoftemo_GB
db_Resolve.H          0.41            -0.08
w_prtnrcollabcomm_GB 0.46            -0.19
w_posexpectns_GB      1.00            -0.19
w_ownsoftemo_GB       -0.19            1.00
Sample size
[1] 257
Probability values (Entries above the diagonal are adjusted for multiple tests.)
      db_Resolve.H w_prtnrcollabcomm_GB
db_Resolve.H      0.00                0
w_prtnrcollabcomm_GB 0.00                0
w_posexpectns_GB    0.00                0
w_ownsoftemo_GB     0.22                0
      w_posexpectns_GB w_ownsoftemo_GB
db_Resolve.H          0                0.22
w_prtnrcollabcomm_GB 0                0.01
w_posexpectns_GB      0                0.01
w_ownsoftemo_GB        0                0.00

To see confidence intervals of the correlations, print with the short=FALSE option
> |
```

Garrett Brownlee
Dr. Kieth Sanford
PSY 4400 01
11/15/2025

Step #6 (w/ regression equation #1):

```
Console Terminal Background Jobs
R 4.5.1 ~ /Brownlee/
> model3 <- lm(data = proj4, db_Resolve.H ~ w_prtnrcollabcomm_GB + w_pospectns_GB + w_ownsoftemo_GB)
> summary(model3)

Call:
lm(formula = db_Resolve.H ~ w_prtnrcollabcomm_GB + w_pospectns_GB + w_ownsoftemo_GB, data = proj4)

Residuals:
    Min       1Q   Median       3Q      Max
-3.5090 -1.2440  0.0425  0.9907  5.0677

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)    1.942630   0.392243   4.953 1.34e-06 ***
w_prtnrcollabcomm_GB 0.020092   0.017940   1.120  0.264
w_pospectns_GB    0.100353   0.017342   5.787 2.12e-08 ***
w_ownsoftemo_GB    0.004344   0.026033   0.167  0.868
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.582 on 253 degrees of freedom
Multiple R-squared:  0.1702,    Adjusted R-squared:  0.1603
F-statistic: 17.29 on 3 and 253 DF,  p-value: 3.024e-10

> |
```

Step #7 (w/ regression equation #1):

```
Console Terminal Background Jobs
R 4.5.1 ~ /Brownlee/
> model4 <- lm(data = proj4, scale(db_Resolve.H) ~ scale(w_prtnrcollabcomm_GB) + scale(w_pospectns_GB) + scale(w_ownsoftemo_GB))
> summary(model4)

Call:
lm(formula = scale(db_Resolve.H) ~ scale(w_prtnrcollabcomm_GB) + scale(w_pospectns_GB) + scale(w_ownsoftemo_GB), data = proj4)

Residuals:
    Min       1Q   Median       3Q      Max
-2.03227 -0.72049  0.02461  0.57378  2.93507

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -1.179e-16   5.716e-02   0.000  1.000
scale(w_prtnrcollabcomm_GB) 7.278e-02   6.499e-02   1.120  0.264
scale(w_pospectns_GB)    3.759e-01   6.496e-02   5.787 2.12e-08 ***
scale(w_ownsoftemo_GB)    9.811e-03   5.879e-02   0.167  0.868
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.9163 on 253 degrees of freedom
Multiple R-squared:  0.1702,    Adjusted R-squared:  0.1603
F-statistic: 17.29 on 3 and 253 DF,  p-value: 3.024e-10

> |
```

Step #8 (w/ regression equation #1):

In a regression analysis, husband-reported conflict resolution was predicted using wives' ratings of partner collaborative communication, wives' positive expectancies, and wives' ratings of their own soft emotion. The three predictors explained a significant portion of variance in husband-reported conflict resolution ($R^2 = .17$, $F(3, 253) = 17.29$, $p < .001$). Correlations and standardized betas are reported in Table 1. There was a medium positive correlation between wives' positive expectancies and husband-reported conflict resolution, and positive expectancies explained unique variance in conflict resolution after controlling for the effects of the other variables in the equation. Although there was a small positive correlation between wives' ratings of partner collaborative communication and conflict resolution, the effect became non-significant after controlling for other variables. Notably, wives' ratings of their own soft emotion were not significantly related to husband-reported conflict resolution.

Table 1.

Predicting husband-reported conflict resolution

	Correlation	Standardized beta
Wife rating of partner collaborative communication	.24***	.07
Wife positive expectancies	.41***	.38***
Wife rating of own soft emotion	-.08	.01

* $p < .05$, ** $p < .01$, *** $p < .001$

Step #9 (w/ regression equation #1):

1. **Screen shots.** This section should show necessary screen shots with the correct commands and results for an analysis (correlations, regression with unstandardized betas, and standardized betas).

Possible points	Deductions	Score
6	0	6

Criterion	Deduction
A. Section does not contain any content errors (missing steps or incorrect results).	3
B. Section contains no more than two content errors.	3
C. Screenshot size and resolution allow it to be read easily.	3
D. Screenshot cropped and does not show command lines with errors.	1
E. Correctly followed instructions to create scales.	3

2. **Written summary.** This section includes a written summary of a dichotomous contrast analysis.

Possible points	Deductions	Score
24	0	24

Criterion	Deduction
A. The summary demonstrates an understanding of the basic results and does not include any major mistakes or misinterpretations (such as a mix up about whether results are significant, a contradictory statement implying	6

	that results are both significant and not significant, or a completely wrong interpretation).	
	B. The first sentence tells the reader what type of analysis is being reported, and specifically uses the word “regression,” names the outcome variable, and states that this outcome is being predicted by (or regressed on) three named predictor variables.	2
	C. Summary reports results pertaining to the omnibus effect of all three predictors. When reporting a result that pertain to all three predictors, the statement uses wording that makes it clear it is talking about all three predictors.	2
	D. Summary reports a correct R^2 value	2
	E. Summary reports a correct F that is correctly formatted with the correct degrees of freedom	2
	F. Summary reports a correct p value	2
	G. Summary refers reader to a table before reporting results from correlations and betas.	2
	H. <i>Results include a variable with a significant correlation and beta</i> , and the summary includes a correct interpretation of the correlation for this variable. The interpretation must clarify direction (positive or negative), clarify size (trivial, small, medium or large), name the two variables, and use the word “correlation.”	2
	I. <i>Results include a variable with a significant correlation and beta</i> , and the summary includes a correct interpretation of the beta for this variable. The interpretation clarifies whether an effect remained significant after controlling for other variables, and it must NOT call the effect a correlation.	2
	J. <i>Results include a variable with a significant correlation but non-significant beta</i> , and the summary includes a correct	2

	interpretation of the correlation for this variable. The interpretation must clarify direction (positive or negative), clarify size (trivial, small, medium or large), name the two variables, and use the word “correlation.”	
	K. <i>Results include a variable with a significant correlation but non-significant beta</i> , and the summary includes a correct interpretation of the beta for this variable. The interpretation clarifies whether an effect remained significant after controlling for other variables, and it must NOT call the effect a correlation.	2
	L. <i>Results include a variable with a non-significant correlation and non-significant beta</i> , and the summary includes a description of these results. The summary should NOT say anything that implies that there is any kind of association between these variables. For example, it should NOT be called a trivial or small effect.	2
	M. The summary is easy to read. All the previous components are in the order listed above or in some other sensible order. The summary does not contain unnecessary information. It does not contain sentences that are difficult to understand.	2
	N. The summary includes a table with a column listing correct correlation values and correctly using asterisks to indicate correlation <i>p</i> -values.	2
	O. The table also includes a column listing correct standardized betas values and correctly using asterisks to indicate beta <i>p</i> -values.	2
	P. The table includes labels for the two columns, names for the variables, and an appropriate note stating the meaning of the asterisks.	2

Garrett Brownlee
 Dr. Kieth Sanford
 PSY 4400 01
 11/15/2025

3. Attachments.	Possible points	Deductions	Score
	0.5	0	0.5

Criterion	Deduction
A. This grading table has been completed and is attached.	0.5

-----Steps 4-9 repeated-----

Step #5 (w/ regression equation #2):

```

Console Terminal Background Jobs
R 4.5.1 - ~/Brownlee/
> listHVA <- c('att_132.H', 'att_133.H', 'att_135.H', 'att_141.H', 'att_142.H', 'att_144.H', 'att_146.H')
> proj4$H_validattrbns_GB <- rowSums(proj4[, listHVA])
> listHPE <- c('exp_152.H', 'exp_153.H', 'exp_154.H', 'exp_155.H')
> proj4$H_posexpect_GB <- rowSums(proj4[, listHPE])
> listHPHE <- c('pem_31.H', 'pem_34.H', 'pem_41.H', 'pem_44.H')
> proj4$H_prtnrhardemo_GB <- rowSums(proj4[, listHPHE])
> library(psych)
> listwsat <- c('dnr_satis.w', 'H_validattrbns_GB', 'H_posexpect_GB', 'H_prtnrhardemo_GB')
> corr.test(proj4[, listwsat])
call:corr.test(x = proj4[, listwsat])
Correlation matrix
      dnr_satis.w H_validattrbns_GB
dnr_satis.w      1.00      0.22
H_validattrbns_GB 0.22      1.00
H_posexpect_GB    0.43      0.43
H_prtnrhardemo_GB -0.11     -0.21
      H_posexpect_GB H_prtnrhardemo_GB
dnr_satis.w      0.43     -0.11
H_validattrbns_GB 0.43     -0.21
H_posexpect_GB    1.00     -0.26
H_prtnrhardemo_GB -0.26      1.00
Sample size
[1] 257
Probability values (Entries above the diagonal are adjusted for multiple tests.)
      dnr_satis.w H_validattrbns_GB
dnr_satis.w      0.00      0
H_validattrbns_GB 0.00      0
H_posexpect_GB    0.00      0
H_prtnrhardemo_GB 0.08      0
      H_posexpect_GB H_prtnrhardemo_GB
dnr_satis.w      0      0.08
H_validattrbns_GB 0      0.00
H_posexpect_GB    0      0.00
H_prtnrhardemo_GB 0      0.00

To see confidence intervals of the correlations, print with the short=FALSE option
> |
  
```

Garrett Brownlee
Dr. Kieth Sanford
PSY 4400 01
11/15/2025

Step #6 (w/ regression equation #2):

```
Console Terminal Background Jobs x
R 4.5.1 · ~/Brownlee/
> model5 <- lm(data = proj4, dnr_satis.W ~ H_validattrbns_GB + H_posexpect_GB + H_prtnrhardemo_GB)
> summary(model5)

Call:
lm(formula = dnr_satis.W ~ H_validattrbns_GB + H_posexpect_GB +
    H_prtnrhardemo_GB, data = proj4)

Residuals:
    Min       1Q   Median       3Q      Max
-11.8035  -1.2963   0.8304   2.0828   7.2450

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)    7.400856    0.987615   7.494  0.000 ***
H_validattrbns_GB  0.028440    0.040149   0.708  0.479
H_posexpect_GB   0.248098    0.038130   6.507  0.000 ***
H_prtnrhardemo_GB 0.009045    0.051265   0.176  0.860
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.193 on 253 degrees of freedom
Multiple R-squared:  0.1893,    Adjusted R-squared:  0.1797
F-statistic: 19.69 on 3 and 253 DF,  p-value: 1.667e-11

> |
```

Step #7 (w/ regression equation #2):

```
Console Terminal Background Jobs x
R 4.5.1 · ~/Brownlee/
> model6 <- lm(data = proj4, scale(dnr_satis.W) ~ scale(H_validattrbns_GB) + scale(H_posexpect_GB) + scale(H_prtnrhardemo_GB))
> model6 <- lm(data = proj4, scale(dnr_satis.W) ~ scale(H_validattrbns_GB) + scale(H_posexpect_GB) + scale(H_prtnrhardemo_GB))
> summary(model6)

Call:
lm(formula = scale(dnr_satis.W) ~ scale(H_validattrbns_GB) +
    scale(H_posexpect_GB) + scale(H_prtnrhardemo_GB), data = proj4)

Residuals:
    Min       1Q   Median       3Q      Max
-3.3476  -0.3676   0.2355   0.5907   2.0548

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)   -1.255e-16  5.650e-02   0.000  1.000
scale(H_validattrbns_GB)  4.472e-02  6.313e-02   0.708  0.479
scale(H_posexpect_GB)    4.166e-01  6.402e-02   6.507  0.000 ***
scale(H_prtnrhardemo_GB)  1.041e-02  5.898e-02   0.176  0.860
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.9057 on 253 degrees of freedom
Multiple R-squared:  0.1893,    Adjusted R-squared:  0.1797
F-statistic: 19.69 on 3 and 253 DF,  p-value: 1.667e-11

> |
```

Step #8 (w/ regression equation #2):

In a regression analysis, wife-reported relationship satisfaction was predicted using husbands' validating attributions, husbands' positive expectancies, and husbands' ratings of partner hard emotion. The three predictors explained a significant portion of variance in wife-reported relationship satisfaction ($R^2 = .19$, $F(3, 253) = 19.69$, $p < .001$). Correlations and standardized betas are reported in Table 2. There was a medium positive correlation between husbands' positive expectancies and wife-reported relationship satisfaction, and positive expectancies explained unique variance in satisfaction after controlling for the effects of the other variables in the equation. Although there was a small positive correlation between husbands' validating attributions and wife-reported satisfaction, the effect became non-significant after controlling for other variables. Notably, husbands' ratings of partner hard emotion were not significantly related to wife-reported relationship satisfaction.

Table 2.

Predicting wife-reported relationship satisfaction

	Correlation	Standardized beta
Husband validating attributions	.22***	.04
Husband positive expectancies	.43***	.42***
Husband rating of partner hard emotion	-.11	.01

* $p < .05$, ** $p < .01$, *** $p < .001$

Step #9 (w/ regression equation #2):

4. **Screen shots.** This section should show necessary screen shots with the correct commands and results for an analysis (correlations, regression with unstandardized betas, and standardized betas).

Possible points	Deductions	Score
6	0	6

Criterion	Deduction
F. Section does not contain any content errors (missing steps or incorrect results).	3
G. Section contains no more than two content errors.	3
H. Screenshot size and resolution allow it to be read easily.	3
I. Screenshot cropped and does not show command lines with errors.	1
J. Correctly followed instructions to create scales.	3

5. **Written summary.** This section includes a written summary of a dichotomous contrast analysis.

Possible points	Deductions	Score
24	0	24

Criterion	Deduction
Q. The summary demonstrates an understanding of the basic results and does not include any major mistakes or misinterpretations (such as a mix up about whether results are significant, a contradictory statement implying	6

	that results are both significant and not significant, or a completely wrong interpretation).	
	R. The first sentence tells the reader what type of analysis is being reported, and specifically uses the word “regression,” names the outcome variable, and states that this outcome is being predicted by (or regressed on) three named predictor variables.	2
	S. Summary reports results pertaining to the omnibus effect of all three predictors. When reporting a result that pertain to all three predictors, the statement uses wording that makes it clear it is talking about all three predictors.	2
	T. Summary reports a correct R^2 value	2
	U. Summary reports a correct F that is correctly formatted with the correct degrees of freedom	2
	V. Summary reports a correct p value	2
	W. Summary refers reader to a table before reporting results from correlations and betas.	2
	X. <i>Results include a variable with a significant correlation and beta</i> , and the summary includes a correct interpretation of the correlation for this variable. The interpretation must clarify direction (positive or negative), clarify size (trivial, small, medium or large), name the two variables, and use the word “correlation.”	2
	Y. <i>Results include a variable with a significant correlation and beta</i> , and the summary includes a correct interpretation of the beta for this variable. The interpretation clarifies whether an effect remained significant after controlling for other variables, and it must NOT call the effect a correlation.	2
	Z. <i>Results include a variable with a significant correlation but non-significant beta</i> , and the summary includes a correct	2

	interpretation of the correlation for this variable. The interpretation must clarify direction (positive or negative), clarify size (trivial, small, medium or large), name the two variables, and use the word “correlation.”	
	AA. <i>Results include a variable with a significant correlation but non-significant beta</i> , and the summary includes a correct interpretation of the beta for this variable. The interpretation clarifies whether an effect remained significant after controlling for other variables, and it must NOT call the effect a correlation.	2
	BB. <i>Results include a variable with a non-significant correlation and non-significant beta</i> , and the summary includes a description of these results. The summary should NOT say anything that implies that there is any kind of association between these variables. For example, it should NOT be called a trivial or small effect.	2
	CC. The summary is easy to read. All the previous components are in the order listed above or in some other sensible order. The summary does not contain unnecessary information. It does not contain sentences that are difficult to understand.	2
	DD. The summary includes a table with a column listing correct correlation values and correctly using asterisks to indicate correlation <i>p</i> -values.	2
	EE. The table also includes a column listing correct standardized betas values and correctly using asterisks to indicate beta <i>p</i> -values.	2
	FF. The table includes labels for the two columns, names for the variables, and an appropriate note stating the meaning of the asterisks.	2

Garrett Brownlee
Dr. Kieth Sanford
PSY 4400 01
11/15/2025

6. Attachments.

Possible points	Deductions	Score
0.5	0	.5

Criterion	Deduction
B. This grading table has been completed and is attached.	0.5